

Conservation Genetics Practice Problems #2

- 1) List three factors that can cause linkage disequilibrium.

- 2) Assume a population is in Hardy-Weinberg Equilibrium at two biallelic loci—A and B—where alleles A_1 and B_1 are on the same chromosome and perfectly associated at time $t = 0$, and alleles A_2 and B_2 are on the same chromosome and perfectly associated at time $t = 0$. After 3 generations of meiosis with a recombination rate of $c = 0.2$, which of the following statements is true?
 - a) allele frequencies have changed but haplotype frequencies have not; evolution has occurred
 - b) allele frequencies have changed but haplotype frequencies have not; evolution has not occurred
 - c) haplotype frequencies have changed but allele frequencies have not; evolution has occurred
 - d) haplotype frequencies have changed but allele frequencies have not; evolution has not occurred

- 3) A sample ($n = 20$) from a population of small wriggly critters is genotyped at biallelic loci A and B , producing the following haplotype counts: $A_1B_1=5$, $A_1B_2=2$, $A_2B_1=3$, and $A_2B_2=5$. Using these data, calculate D .

- 4) Why is the “equilibrium” in linkage disequilibrium different than the “equilibrium” in Hardy-Weinberg Equilibrium?

- 5) A sample ($n = 10$) of a population of large sedate critters is genotyped at biallelic loci A and B , producing the following individual genotypes: A_1A_1/B_1B_1 (3), A_1A_2/B_1B_2 (3), A_2A_2/B_1B_1 (1), A_1A_2/B_2B_1 (3). Evaluate the following statements:
 - a) This population is in Hardy-Weinberg Equilibrium at locus A (**T/F**)
 - b) This population is in Hardy-Weinberg Equilibrium at locus B (**T/F**)
 - c) Alleles at locus A and B assort independently (**T/F**)